

LLM: Tokenizer & Dataset Builder

Tokenization ni eng alo usullaridan biri deb BPE tokenization ni aytganmiz.

1. Subword BPE: sequence ni uzunligini kattaytirib yubormaydi. <unk> token problemasini yuqotadi. LLM ning kirish qismida va modelning asosiy vocabulary.

Usul	Lugat	Tokenlar	Token/belgi
character	79	94,501	1.000
word	3,280	15,045	0.159

Faqat 1 marta uchragan so'zlar: 1,884 / 3,280 (57.4%)

Eng kam uchraydigan 10 ta so'z: ['variety.', 'Analyze', 'trainer', 'perspectives', 'insights.', 'urine', 'color', 'status.', 'electrolytes', 'impairs']

2. BPE Merge rules: eng kup duplicate bulodigon token yokida belgilar juftligini birlashtirish va ularning ketma-ketli tartibini saqllovchi qoidalar tuplami. Nega bu kerak desak, tokenizator matnni har doim bir hil mantiqiy tartibda encode qilish uchun juda kerak.

```
Matn: low low low low low lower lower newest newest newest newest newest newest widest widest widest

merge 1: 'e' + 's' -> 'es' (uchradi: 9)
merge 2: 'es' + 't' -> 'est' (uchradi: 9)
merge 3: 'est' + '_' -> 'est_' (uchradi: 9)
merge 4: 'l' + 'o' -> 'lo' (uchradi: 7)
merge 5: 'lo' + 'w' -> 'low' (uchradi: 7)
merge 6: 'n' + 'e' -> 'ne' (uchradi: 6)
merge 7: 'ne' + 'w' -> 'new' (uchradi: 6)
merge 8: 'new' + 'est_' -> 'newest_' (uchradi: 6)
merge 9: 'low' + '_' -> 'low_' (uchradi: 5)
merge 10: 'w' + 'i' -> 'wi' (uchradi: 3)
```

3. Vocabulary size trade-offs: tokenizer vocabulary sizeini kichik (4k) yokida katta (32k) orasida balance tanlash va uni modelga tasiri. Kichik model embedding ni kichkinaytiradi lekin compute harajatini oshiradi. Katta vocabulary esa sequenceni uzunligini kichkinaytiradi.

vocab	embed params	ulush (30M da)	
1,000	384,000	1.3%	
4,000	1,536,000	5.1%	
8,000	3,072,000	10.2%	<-- bizniki
16,000	6,144,000	20.5%	
32,000	12,288,000	41.0%	
50,257	19,298,688	64.3%	
65,535	25,165,440	83.9%	

50257 - GPT-2 ning lug'ati. 65535 - uint16 dagi chegara.

4. **Special Tokens**: modeldagi matn chegaralari yoki **mahsus holatlarni tushuntirish** uchun ishlatiladigan **mahsus** hizmatchi **tokenlar**. <eos> kabi tokenlarda yani, end of sequences ni bildiradi.

```
Tokenizer o'rgatiladigan hujjatlar: 170  
O'rgatish vaqti: 0.1 soniya  
Saqlandi: C:\Users\User\Desktop\ML_Lesson\Projects\LLM\trainerAI_LLM\our-llm\tokenizer\tokenizer.json  
Fayl hajmi: 0.19 MB  
Haqiqiy vocab_size: 3106
```

5. **Round-trip test**: ID ga encode qilingan model – decode qilsa aynan asl matnga qaytishini tekshirish. **Encoded = Decoded**. Bu test dan albatta utish kerak bumasam tokenizer yoki boshqa muammo mavjud

```
--- Chekka holatlar ---  
[OK ] 8 token 'lorem ipsum dolor'  
[OK ] 10 token 'hello my name is..'  
[OK ] 14 token 'emoji: 🤖🚀'  
[OK ] 14 token '你好, éàü'  
[OK ] 19 token " bir nechta bo'sh joy "  
[OK ] 9 token 'row1\nrow2\nrow10'  
[OK ] 12 token '\ttab bilan boshlandi'  
[OK ] 17 token '46239847 va 3.14159'  
[OK ] 0 token ''  
[OK ] 12 token 'Hiiiiiiiiiii'  
  
10/10 o'tdi
```

Dataset Builder

Document-level data splitting: malumotlarni train va validation guruhlariga qator darajada emas balki, butunlay **document darajasida ajratish**. **Data leakage oldini olish**.

```
Train hujjatlar : 169  
Val hujjatlar   : 1 (0.59%)  
  
Split toza: kesishuv yo'q, hujjat yo'qolmadi.
```